moving to lossless compression techniques and FLAC stands as a good open source option. The compression ratio varies from 0 to 8, with the best quality standing at a compression ratio of 0, which reproduces the original sound stream. With a compression ratio of 8, there is little trade-off between the size and quality. The beauty of FLAC stands out when using the audio stream for further processing. When audio is recorded along with a video stream in FLAC or OGG, the audio stream can be further processed and modified, without any loss in the process. The trade offs with using FLAC is in the file size, which are often double that of rival file formats and the rate of compression is very slow as the file format tries to preserve all the data while reducing the size. Hence, the decompression rates are slow, and the digital audio players would need to use more storage and processing power to process FLAC files.

flac is the preferred choice for audiophiles looking for quality

AAC

AAC or Advanced Audio Coding is another popular audio file format, mainly popularized by Apple with its use in the downloaded music from iTunes and in Apple Music. AAC promises to deliver better audio quality compared to MP3 and is the default standard for YouTube, iPhone, iPad, Nintendo Devices, Playstations and a few smartphone manufacturers. Jointly developed by Bell Labs, Fraunhofer Institute, Dolby Labs, Sony and Nokia, AAC was pitched to overtake MP3 in terms of usage by focusing on its free-to-use license. Major improvements over MP3 include increase in the sampling frequencies and fixes to the usage of variable bitrates. There were also improvements in the capacity to handle high audio frequencies.

WMA

Windows Media Audio (WAV), as the name suggests is a proprietary codec developed by Microsoft. It was developed to work with Microsoft programs like Windows Media Player and Xbox Live services. It provided an avenue for Microsoft to introduce DRM on its now defunct Zune and other ventures